USER MANUAL · REV 1.0

BIG MUFF

Quad-Preset MIDI Fuzz · a faithful clone of the Big Muff-style fuzz (Green Russian era) · handmade in Austria by Walter Craffonara

NOTE This pedal is a faithful, circuit-for-circuit clone of the Big Muff-style fuzz of the Green Russian era, expanded with functions I added: a relay-switched quad-preset system and MIDI. The name “Big Muff” is used only to identify the original circuit this build recreates. CRAFFONARA FX is an independent builder and is not affiliated with, authorised by or endorsed by Electro-Harmonix / New Sensor Corp.

1 Overview & Concept

At its core this is a faithful clone of the Big Muff-style fuzz of the Green Russian era — famous for its thick, saturated wall of fuzz. The clipping section can be switched between full fuzz and a tighter overdrive voicing.

What I added is the part the original never had: the Fuzz and Tone controls exist twice. A relay silently switches between the two pots of each channel — by soft-touch button on the pedal, or remotely via MIDI. Dial in a rhythm sound and a lead sound and recall both hands-free.

2 Controls & Connections

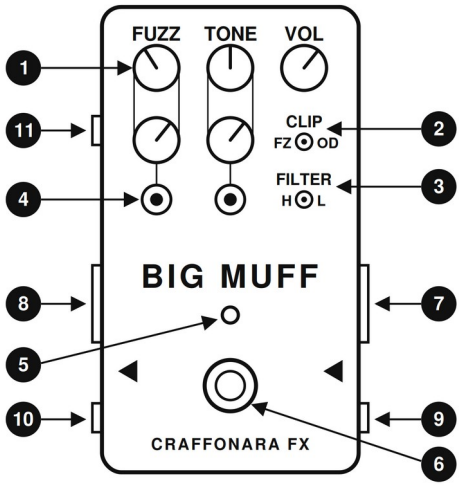


Fig. 1 — Panel and connections

1 FUZZ · TONE · VOL the main controls for fuzz amount, tone colour and output level. FUZZ and TONE each exist twice: an upper pot with a white marker and a lower pot with a red marker.

2 CLIP toggle FZ = full Green-Russian fuzz, OD = a tighter, more focused overdrive voicing with less compression.

3 FILTER toggle H = high peak for an aggressive voicing, L = low peak for a smoother, darker response.

4 PRESET buttons one soft-touch button per channel (FUZZ, TONE), each driving its own relay. LED ring dark = preset #1 (upper pot, white marker); LED ring lit = preset #2 (lower pot, red marker).

5 STATUS LED shows whether the effect is engaged or bypassed. It does NOT indicate which preset is active — that is shown by the LED ring around each PRESET button.

6 FOOTSWITCH silent soft-touch switch, engages / bypasses the effect (relay true bypass).

7 INPUT 6.35 mm mono jack from your guitar or the previous pedal.

8 OUTPUT 6.35 mm mono jack to your amp or the next pedal.

9 MIDI IN TRS socket for incoming MIDI (Type A).

10 MIDI THRU TRS socket passing incoming MIDI on to the next device.

11 POWER 9 V DC, center-negative.

3 The Quad-Preset System

Fuzz and Tone each exist twice on the panel: an upper pot with a white marker and a lower pot with a red marker. Each channel has its own PRESET button (4), and each button drives its own relay — the two relays work completely independently of one another.

The LED ring around a PRESET button shows which pot that channel is using:

LED ring dark (preset #1) the upper pot with the white marker is active; the lower pot is muted.

LED ring lit (preset #2) the lower pot with the red marker is active; the upper pot is muted.

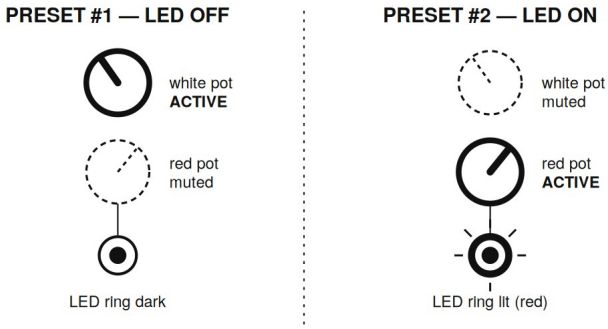


Fig. 2 — Preset states of one channel (Fuzz or Tone)

Because the two relays switch independently, the two channels give you 2 × 2 = four complete settings in one pedal — a quad-preset system. Recall any of the four by PRESET button or over MIDI: for example a rhythm sound, a lead sound, and two in-between voicings.

> **NOTE** Preset switching is relay-based and click-free. Only the selected pot sits in the signal path, so the muted pot causes no tone loss.

4 Voicing Controls

CLIP toggle — FZ / OD FZ gives the full, compressed Green-Russian fuzz wall; OD tightens the clipping into a more focused overdrive with more note definition.

FILTER toggle — H / L sets the tone-stack peak. H is aggressive and cuts through a dense mix; L is smoother and darker.

5 MIDI Reference

The pedal responds to MIDI Control Change (CC) messages on its MIDI channel (see the next section).

CC	Value	Command
00	00	TONE preset #1 and FUZZ preset #1
	01	TONE preset #2 and FUZZ preset #1
	02	TONE preset #1 and FUZZ preset #2

CC	Value	Command
	03	TONE preset #2 and FUZZ preset #2
10	00	TONE preset #1
	01	TONE preset #2
	02	Toggle TONE preset #1 / #2
20	00	STATE: OFF — effect bypassed
	01	STATE: ON — effect engaged
30	00	FUZZ preset #1
	01	FUZZ preset #2
	02	Toggle FUZZ preset #1 / #2

6 Setting the MIDI Channel

1. Disconnect the pedal from the power supply.
2. Press and hold the FOOTSWITCH while connecting the power supply — the STATUS LED lights up and starts to flash, showing the pedal is in channel-set mode.
3. Press the FOOTSWITCH as many times as the channel number you want to set (e.g. three times for MIDI channel 3). The STATUS LED flashes back with each press.
4. Press and hold the FOOTSWITCH until the STATUS LED goes dark and the pedal switches off completely.
5. On the next power-up the pedal responds to the new MIDI channel.

7 Power

Run the pedal on 9 V DC, center-negative, from any regulated pedalboard supply. There is no battery compartment.

8 Specifications

Parameter	Value
Type	Fuzz — faithful clone of the Big Muff-style circuit (Green Russian era), with a quad-preset system and MIDI
Controls	FUZZ ×2, TONE ×2 (relay-switched presets), VOL, CLIP toggle, FILTER toggle
Clipping	FZ = full fuzz, OD = tighter overdrive voicing
Presets	2 × Fuzz, 2 × Tone — relay-switched, four combinations
Bypass	Relay true bypass, silent soft-touch footswitch
MIDI	IN + THRU (TRS, Type A), Control Change
Audio I/O	6.35 mm mono input and output
Power	9 V DC, center-negative
Enclosure	Brushed aluminium, CNC-drilled (Hammond 1590B format)

NOTE Trademark notice: “Big Muff” and “Electro-Harmonix” are trademarks of New Sensor Corporation. They are used here purely to describe the original circuit that this independent build recreates. CRAFFONARA FX is not affiliated with, authorised by or endorsed by the trademark holder.